Understanding the Problem

I approached understanding the challenge by identifying that the goal was to read a file and count how many vowels it contains. This meant I needed to work with file input and analyze text character by character.

At first, I was confused because the program creates a file object but also asks the user for a file name. I realized that the actual file being used is the one entered by the user.

```java
                textFile = new File(fileName);
```

```java
        System.out.print("Enter the file name: ");
        fileName = scanner.nextLine();
```

Planning the Solution

Before coding, I broke the problem into steps: getting input, reading the file, looping through text, checking vowels, and counting them.

I chose `BufferedReader` to read the file line by line and used a string `"aeiou"` to check for vowels.

Code reference:

```java
        try (BufferedReader reader = new BufferedReader(new FileReader(fileName))) {
```

```java
                        if ("aeiou".indexOf(ch) != -1)
                        {
                            vowelSum++;
                        }
```

Implementation

I built the program step by step. First I handled input, then file reading, and finally the logic for counting vowels.

I tested the program by running it with a sample text file and checking if the output matched the expected number of vowels.

```java
            // convert to lower case
            lowercaseText = lineInFile.toLowerCase();

            // iterate through text
            for (int i = 0; i < lowercaseText.length(); i++)
            {
                char ch = lowercaseText.charAt(i);

                // check if vowel
                if ("aeiou".indexOf(ch) != -1)
                {
                    vowelSum++;
```

Overcoming Challenges

The most difficult part was understanding how file reading works and how to properly loop through each character in the file.

When I got stuck, I focused on one part at a time, like first making sure the file was being read, then adding the vowel-checking logic.

Code reference:

```java
        catch (IOException e)
        {
            System.out.println("Error reading file: " + e.getMessage());
        }
```

Learning

I learned how to read files using `BufferedReader`, how to loop through strings, and how to check characters efficiently.

This will help me in future programs that involve file handling and text processing.

Code reference:

```java
            System.out.println("Number of vowels: " + vowelCount);
```